



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
BANGLADESH UNIVERSITY OF BUSINESS AND TECHNOLOGY

COURSE OUTLINE

1	Faculty	Faculty of Engineering & Applied Sciences (FEAS)
2	Department	Department of Computer Science and Engineering (CSE)
3	Program	Bachelor of Science in Computer Science and Engineering (B.Sc. in CSE)
4	Course Code	EEE 212 (BNQF Code: 0612212)
5	Course Title	Electronic Devices and Circuits Lab
6	Course Type	Interdisciplinary Course
7	Pre-requisites & Co-requisites	EEE 102 (BNQF-0612202) EEE 101 (BNQF-0612201)
8	Credit Hours/Value	1.5
9	Contact Hours	17 weeks × 2 classes per week × 1.25 hours per class = 42.5 total hours
10	Year-Semester	1-2
11	Academic Session	Spring 2025

12	Class Schedule	Intake – Section (Shift)	Class Day	Class Hours	Venue
		53-1 (Day)	Sunday	10:30 AM-01:00 PM	Room: 505 (B-3)
13	Course Website	Google Classroom Code: k5ai225 Meet Link: https://meet.google.com/dud-aoyr-wgy			

14	Course Teacher's Information	Name (Code): Md. Ashik Khan (MDAK) Designation: Lecturer, Department of EEE	Specialization: Control System, Converter, Power electronics, Electronics Email: mdashikkhan@bubt.edu.bd Cell No. 01885539040
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15	Counselling Hour/Tutorial	Day	Counseling Hours	Venue
		Tuesday	02.45 PM to 05.15 PM	B2/711

16	Accessories & Aids	Students must carry learning materials like lecture notes, calculator, pen, pencil, eraser etc. in the classroom. Borrowing learning materials in the classroom or exam room from fellow students is prohibited. A student is also advised to keep a separate class note (khata) of 50 pages for the course during class hours.
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17	Course Rationale	This course will equip the students with hands on and experimental skills in analog electronics and devices. To focus on ongoing technological developments, this course envisions to investigate the fundamental working principle and features of different types of diodes, Bipolar Junction transistors (BJT), different types of Field Effect Transistors (FETs) and Operational Amplifiers (Op-Amp) along with the applications in analog electronics.
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18	Course Objective	The Course has therefore been designed to: Obj.1. Teach the students to investigate the performance of different analog electronic devices. Obj.2. Apply basic concepts of electronics for analyzing analog electronic circuits along with graphical phenomena. Obj.3. Introduce the students in designing basic analog electronic devices.
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19	Course Learning Outcomes (CLOs)	Upon completing this course, students will be able to: CLO1: Investigate the performance of different digital devices, its characteristics and graphical analysis in electronic circuits. (K1, K2 & K3) CLO2: Design and develop of different electronic devices using Diode, BJT, MOSFET and op-amp for the procedure of amplification, rectification, mathematical operation, filtering and oscillation. (K1, K2 & K3; P1) CLO3: Select and use modern hardware and software tools and devices. (K1, K2 & K3; P1) CLO4: Assess societal, health, safety, legal and cultural issues involved with the mini project. (K7) CLO5: Realize the impact of societal, environmental and sustainable development issues for the design solution of mini project. (K7) CLO6: Apply professional ethics and responsibilities in the implementation of mini project. (K7) CLO7: Work individually and in a team. CLO8: Communicate and share knowledge, data, information, results etc. with others. CLO9: Apply engineering project management knowledge and skill to implement the mini project. (K6; P1) CLO10: Gather and apply knowledge, data and information from various multidisciplinary sources to analyze, design and implement the mini project. (K5; P1)
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20	Mapping of CO – PO – K, P/A – Learning Domain, and Method of Delivery and Assessment Tool					
	CLOs	POs & CF	Learning Domain	Ks & Ps**	Delivery Methods / Activities	Assessment Tools
	CLO1	PLO3 CF=2	Cognitive / Understanding	K1, K2, K3	- Classroom Lecture (PPP &/or WBT*) - Classroom Discussion & Exercise practice - Analysis and design problem solving - Course Counseling	Indirect: In class response, course counseling, course end survey Direct: Class participation and activity, class test, assignment, midterm and final examinations
	CLO2	PLO4 CF=2	Cognitive / Analyzing	K1, K2, K3; P1		
	CLO3	PLO5 CF=2	Cognitive / Analyzing	K1, K2, K3; P1		
	CLO4	PLO6 CF=2	Cognitive/ Evaluating	K7		
	CLO5	PLO7 CF=2	Affective	K7		
	CLO6	PLO8 CF=2	Affective	K7		

CLO7	PLO9 CF=2	Psychomotor	---		
CLO8	PLO10 CF=2	Affective	---		
CLO9	PLO11 CF=2	Psychomotor	K6; P1		
CLO10	PLO12 CF=2	Psychomotor	K5; P1		
*PPP: Power Point Presentation & WBT: White Board Teaching					
**Ks: Knowledge Profiles & Ps: Profiles of Complex Engineering Problem					
Correlation of COs to POs					
Correlation Criteria			Correlation Level	Correlation Factor (CF)	
Less than 25%			Almost no correlation	0	
Equal to or greater than 25% and below 50%			Poor (Low)	1	
Equal to or greater than 50% and below 75%			Moderate (Medium)	2	
Equal to or greater than 75% and up to 100%			Significant (High)	3	

21	Teaching-Learning Strategy	The course's teaching-learning process is designed to achieve its intended learning outcomes. Various classroom tools, such as multimedia projectors with desktop computers, whiteboards, and overhead projectors, are used to make the process engaging, effective, and comprehensive. The primary method of teaching is through classroom lectures, wherein most of the course content is covered in the lecture notes. For the remaining topics, textbooks are utilized, with additional references provided for students to study on their own. Lecture materials are posted on Google Classroom web pages and also provided as hard copies in the classroom. To ensure that students achieve the expected performance and knowledge level, classroom discussions, PowerPoint presentations, problem-solving using whiteboard markers, and homework or home studies are used. Counseling is also offered to help students with weak areas. Formative assessments of individual students are done through inside and outside classroom discussions, in-class eye contact and clicker questions, homework, and students' responses. A course-end survey is also conducted. Summative assessments are done through class participation and performance observation, assignments, class tests, and semester midterm and final examinations. If a student is absent from a class for any reason, they are advised to do self-study and take tutorials from the class teacher to make up for the missed class. Supplementary examinations are available for students who missed the midterm or final examinations due to valid reasons. These supplementary exams are more challenging than the regular exams.
22	Course Plan	This course consists of 150.0 min./week of class contact hours and an additional 90 min./week of counseling hours to explain students' design problems, provide reading materials, and assist in understanding lecture materials for preparing examinations, class tests, and assignments.

Week No.	Lecture No.	Lecture Topic (syllabus covered)	Lecture Note and Text Book	Teaching-Learning Strategy	Assessment Strategy	CLO
1	1	Setting Context of Teaching and Learning Based on OBE Model	Course profile Hand notes and Text [1] Lab sheet 01	Delivering presentation-based instruction, Delivering lecture-based instruction, classroom discussion, Lab Experiments, group discussion and hands-on lab practice	Continuous Lab performance (CLP), Midterm Lab assessment	CLO1
	2	Exp. No. 01: Introduction to Proteus/P-Spice/Multisim.				
2	3	Exp. No. 02: Study of Diodes' I – V Characteristics.	Text [1] Lab Sheet 02	Delivering lecture-based instruction, classroom discussion, Lab Experiments, group discussion and hands-on lab practice	Continuous Lab performance (CLP), Midterm Lab assessment	CLO1
	4					
3	5	Exp. No. 03: Study of Clipper Circuit by Using Diode.	Text [1] Lab sheet 03	Same as before	Same as before	CLO1
	6					
4	7	Exp. No. 04: Study of Clamper Circuit by Using Diode.	Text [1] Lab Sheet 04	Same as before	Same as before	CLO1
	8					
5	9	Exp. No. 05: Study of Diode Bridge Half Wave Rectifier.	Text [1] Lab Sheet 05	Same as before	Same as before	CLO2
	10					
6	11	Exp. No. 06: Study of Diode Bridge Full Wave Rectifier.	Text [1] Lab Sheet 06	Same as before	Same as before	CLO2
	12					
7	13	Midterm Lab Assessments		problem solving	Midterm Lab assessment	CLO3, CLO7, CLO8
	14	Problems solving based on students' feedbacks				
Midterm Examination (according to the Academic Calendar- and examination schedule)						
8	15	Exp. No. 07: DC Characteristics of BJT (Bipolar Junction Transistor) in Common Emitter.	Text [1] Lab Sheet 07	Delivering lecture-based instruction, classroom discussion, Lab Experiments, group discussion and hands-on lab practice	Continuous Lab performance (CLP), Final Lab assessment	CLO2
	16					

	9	17	Exp. No. 08: DC Characteristics of BJT (Bipolar Junction Transistor) in Common Base.	Text [1] Lab Sheet 08	Same as before	Same as before	CLO2
		18					
	10	19	Exp. No. 09: Application of BJT (Bipolar Junction Transistor) as switch.	Text [1] Lab Sheet 09	Same as before	Same as before	CLO2
		20					
	11	21	Exp. No. 10: Application of Op-amp as inverting, non – inverting, summing amplifier.	Text [1] Lab Sheet 10	Same as before	Same as before	CLO2
		22					
	12	23	Exp. No. 11a: Application of Op-amp as differentiator and integrator.	Text [1] Lab Sheet 11	Same as before	Same as before	CLO2
		24					
	13	25	Exp. No. 11b: Application of Op-amp as active high pass and low pass filters.	Text [1] Lab Sheet 11	Same as before	Same as before	CLO2
		26					
	14	27	Exp. No. 12: I-V Characteristics of N channel enhancement type MOSFET.	Text [1] Lab Sheet 12	Same as before	Same as before	CLO2
		28					
	15	29	Mini Project Presentation			Mini Project assessment	CLO3- CLO10
		30					
	16	31	Brain Storming Week: To provide opportunities to the students to generate new ideas in areas covered in the course		Classroom discussion and problem solving		
		32					
	17	33	Final Lab Assessments			Final Lab assessment	CLO3, CLO7, CLO8
		34					
Final Examination (according to the Academic Calendar and examination schedule)							
23 i	Class Note		1. Lab Manual of the course of Electronic Devices and Circuits Lab for the students of B.Sc. in Computer Science and Engineering (CSE) Program, BUBT by Md. Alamgir Kabir				
ii	Text Books		1. Electronic Devices and Circuit Theory by Robert Boylestad and Louis Nashelsky 2. Microelectronic Circuits by Adel S. Sedra and Kenneth Carless Smith 3. Electronic Devices and Circuits by David A. Bell 4. Electronic Principles by Albert Malvino and David Bates 5. Principles of Electronics by V. K. Mehta 6. Fundamentals of Electric Circuits by Charles Alexander and Matthew Sadiku 7. Op-Amps and Linear Integrated Circuits by Ramakant A. Gayakwad				

24	Assessment and marks distribution criteria	An integral part of this course is engagement in class activities, outside classroom discussions and communications through net and phone. Students may fail the course if they do not participate in the class regularly, class tests and/or complete assignments. Students must actively participate in classroom discussions and complete all sets of works at a satisfactory level, as mentioned in this course outline. The course-specific expectations for students are achieved if				
		75% of students in a section attend more than 70% of classes (determined by summative assessment). - Their active participation in the classroom discussion is targeted up to 80% of total attendee (determined by formative assessment such as eye-contact, clicker questions and group discussions). - Equal or more than 40% of course outcomes must be achieved by the students (summative assessment). - The level of engagement in the studies such as preparing regularly class lectures, class tests, assignments must be more than 60% (formative assessment). - Expected level of participation in the outside class discussion (once in a week, more than 30% of students in a section) by course counselling and using social media like Google Classroom, email and phone call etc. (formative assessment). - Students are assessed according to their individual performance in the examinations, class tests, assignments, and class participations. The final mark calculation and course outcome assessment are done based on the following mark distribution (criteria):				
		Assessment tool	Conducting Number	Mark distribution (%)		
		Class participation and activity	17	05		
		Lab Report	12	10		
		Mini Project Presentation and Viva	1	25		
		Continuous Lab Performance (CLP)	1	30		
		Lab Final	1	30		
		Total Mark		100		
		Class Participation & Activity Performance Criteria				
		Performance level		Mark distribution (%)		
		90% -100%		05		
		80% - 89%		04		
		70% - 79%		03		
		60% - 69%		02		
		50% - 59%		01		
		Less than 50%		Not allowed to sit for the final examination.		
		25	Assessment Strategy			
			Assessment tool	Content, Length and Criteria	Weight (mark)	Due date
Lab Report	Composed of lab reports as provided format for each		10	On each lecture, lab		

	experiment. Lab reports have to be submitted on each lecture and the assigned full mark is 10 for 12 lab reports. A Lab Report must be submitted according to the following format: 1. Experiment No. 2. Name of the Experiment 3. Objectives of the Experiment 4. Theory 5. Equipment List 6. Circuit Diagram & Design Procedure 7. Data Collection 8. Results (Include graph, drawing sheet; if available) 9. Discussions 10. References		reports have to be submitted.
Mini Project Presentation and Viva	Mini project would be assigned to a group consisting of 5/6 members. These mini projects are based on the knowledge and expertise acquired from the theory and lab course. New ideas related the course could also be assigned to the students under mini project segment. To evaluate mini project, a report has to be submitted and a presentation session will be conducted. The assigned full mark for this segment is 25 for 1 mini project. A Mini Project Presentation content must contain the following: 1. Title Page (Project title, Student(s) name and ID with department name & BUBT logo) 2. Background, Motivation and Objectives 3. Literature Review 4. Methodology (Block diagram) 5. Detailed Circuit Diagram 6. Simulation / Hardware Prototype 7. Results and Discussions 8. Conclusion A Mini Project Report must be submitted according to the following format: 11. Project Title, Student(s) name and ID with department name & BUBT logo 12. Introduction 13. Background, Motivation and Objectives 14. Literature Review 15. Methodology 16. Proposed method 17. Simulation / Hardware Prototype 18. Results and Discussions 19. Socio-economic Impact 20. Environmental impact & Sustainability 21. Conclusion 22. References	25	On the 29th lecture, mini project presentation and viva will be conducted.
Continuous Lab	Continuous performance during conducting experiments in	30	On each lecture,

	Performance (CLP)	lab classes would be measured by the course teacher. The evaluation could be based on the understanding the basics for the experiment, work capability and contribution in a group and overall behavior of students at classroom. The assigned full mark for this segment is 30 for 12 experimentations.									continuous lab performance evaluation will be conducted.		
	Lab Final	Lab Final exam will be conducted at the lab class based on the experiments completed at this course. Course teacher can evaluate student for his hands on expertise in the lab course by conducting this exam. The assigned full mark for this segment is 30.								30	After 32 nd lecture, lab final exam will be conducted.		
26	CLO Assessment Criteria	Assessment of CLOs											
		Assessment Area	CO										Assessment Area Mark
			CO 1	CO 2	CO 3	CO 4	CO 5	CO 6	CO 7	CO 8	CO 9	CO 10	
		Attendance and Performance	-	-	-								-
		Lab Report	3		3					4			10
		Mini Project Presentation and Viva	2	2	2	1	1	1	2	10	1	2	24
		Continuous Lab Performance (CLP)		8	8				7	8			31
		Lab Final	30										30
		Total Mark	35	10	13	1	1	1	9	22	1	2	95
27	Rubrics (Attainment Criteria)	CLOs (Taxonomy domain)	Not attained /Failed (0-39%)	Poor (40%-49%)	Moderate (50%-64%)	Good (65%-79%)	Excellent (80%-100%)						
		CLO1 (understand) CLO2 (analyze) CLO3 (analyze) CLO4 (Evaluate) CLO5 CLO6 CLO7 CLO8 CLO9 CLO10	The question is answered with serious deficiencies in understanding and explanation. Applicable method is not almost touched.	The question is answered inadequately by touching the applicable method or without explanations. As a result, a few steps of problem-solving procedure or concepts are not developed properly or missing.	The question is answered partially correct by applying the asking method or concepts, but a few important details are missing.	The question is answered correctly but briefly missing some portions of the important explanation by applying the required method or concepts.	The question is answered correctly with detail explanations using the asking method of solving the problem or concepts with adequate explanation.						

28	Feedback	All kinds of feedback to the students will be produced within a week after the day of holding a class test and midterm examination. No answer script will be shown for the final examination if not challenged by a student. Online/email queries will also be responded within three days by email.			
29	Grading Policy	Letter grades and grade points are used to evaluate the performance of a student in the course:			
		Marks Range	Letter Grade	Grade Point	
		80% and above	A+	:A Plus (Outstanding)	4.00
		75% to less than 80%	A	:A regular (Excellent)	3.75
		70% to less than 75%	A-	:A minus (Very good)	3.50
		65% to less than 70%	B+	: B Plus (Good)	3.25
		60% to less than 65%	B	: B regular (Average)	3.00
		55% to less than 60%	B-	:B minus (Below average)	2.75
		50% to less than 55%	C+	: C Plus (Poor)	2.50
		45% to less than 50%	C	:C regular (Very poor)	2.25
		40% to less than 45%	D	: Pass marginally	2.00
		Less than 40%	F	: Fail	
			I	: Incomplete	
			R	: Retake	
	W	: Withdraw			
30	Additional Course Policies	Assignment	All assignments have equal weights. No late submission will be accepted. Any kind of copied information without proper citation (i.e., plagiarism) in the assignments or any other work is prohibited and treating as a serious academic misconduct which is prosecuted through the Discipline Committee of BUBT. All copied/plagiarized assignments/reports/test papers will be canceled, and the isolated student must prepare a new assignment/project again. Zero tolerance will be shown in this regard. Feedback after submission of long assignments will be provided on hand or by email within two weeks.		
		Class Test	There will be conducted two class tests (CTs) with one quiz or three CTs. All class tests/quizzes have equal weightage. Both regular and surprise CTs/quizzes can be conducted.		
		Closed book assessments	Quizzes, CTs, midterm and final examinations are closed book assessments. Mobile phone is prohibited in the examination hall. Students are insisted on carrying simple scientific calculators to solve the complex calculations and a wrist-watch to follow time during the exam hours.		
		Test Policy	If a student is absent from class test anyway and has made no report to the class teacher personally beforehand, his/her score for that test will be zero. No make-up for the class test will be allowed. No supplementary for midterm and final examinations will be entertained without physical presence and recommendation of the guardian, along with written permission of the department. Supplementary examination questions are much harder than the regular examination questions; therefore, students are discouraged from taking supplementary examinations.		
31	Additional Information	Academic Calendar Fall 2023: https://www.bubt.edu.bd/Home/page_details/Academic_Calender Academic Rules: https://www.bubt.edu.bd/Home/page_details/Rules and Regulations			

	Grading & Evaluation: https://www.bubt.edu.bd/Home/page_details/Evaluation_Grading_System Rules & Regulations: https://www.bubt.edu.bd/Home/page_details/Office_of_the_Registrar																						
32	Bloom's Taxonomy for Teaching-Learning Bloom's Taxonomy is a set of three hierarchical models used to classify educational learning objectives into levels of complexity and specific quality. The three taxonomy domains for achieving learning objectives are cognitive, affective, and psychomotor. Cognitive domain is in the primary focus of educating and frequently used to structure curriculum learning objectives and achieve the level of learning. The three domains and their respective levels are illustrated below:																						
	<table> <tr> <th>Cognitive [C] (Knowledge-based)</th><th>Affective [A] (Emotion-based)</th><th>Psychomotor [P] (Action-based)</th></tr> <tr> <td>Remember</td><td>Receive</td><td>Imitate</td></tr> <tr> <td>Understand</td><td>Respond</td><td>Manipulate</td></tr> <tr> <td>Apply</td><td>Value</td><td>Precision</td></tr> <tr> <td>Analyze</td><td>Organize</td><td>Articulation</td></tr> <tr> <td>Evaluate</td><td>Characterize</td><td>Naturalization</td></tr> <tr> <td>Create</td><td>--- --- ---</td><td>--- --- ---</td></tr> </table>	Cognitive [C] (Knowledge-based)	Affective [A] (Emotion-based)	Psychomotor [P] (Action-based)	Remember	Receive	Imitate	Understand	Respond	Manipulate	Apply	Value	Precision	Analyze	Organize	Articulation	Evaluate	Characterize	Naturalization	Create	--- --- ---	--- --- ---	
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33	Descriptions of Cognitive Domain (Anderson and Krathwohl's updated Taxonomy in 2001): The cognitive domain involves the development of our mental skills and the acquisition of knowledge.																						
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Create	Putting elements together to form a coherent or	Create, invent, compose, predict, plan,																					

	(C6)	functional whole; reorganizing elements into a new pattern or structure through generating, planning, or producing. Creating requires users to put parts together in a new way, or synthesize parts into something new and different creating a new form or product. This process is the most difficult mental function in the new taxonomy.	construct, design, propose, devise, and formulate
34	Graduate Attributes (Program Learning Outcomes) of B.Sc. in EEE Program based on Washington Accord Program Learning Outcomes (PLOs) are brief statements that describe what students are expected to know and be able to do by the time of graduation. These relate to the knowledge, skills, and attitudes that students acquire throughout the entire course of a program. The students of the B.Sc. in CE program are expected to achieve the following graduate attributes or program outcomes at the time of graduation: PLO1: (Engineering knowledge): Apply knowledge of mathematics, natural science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems. PLO2: (Problem analysis): Identify, formulate, and analyze complex engineering problems reaching substantiated conclusions using the first principles of mathematics and natural and engineering sciences. PLO3: (Design/development of solutions): Design solutions for complex engineering problems and design systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations. PLO4: (Investigation): Conduct investigations of complex problems using research-based knowledge and research methods that include the design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions. PLO5: (Modern tool usage): Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling, to complex engineering problems, with an understanding of the limitations. PLO6: (The engineer and society): Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex engineering problems. PLO7: (Environment and sustainability): Understand and evaluate the sustainability and impact of professional engineering work to solve complex engineering problems in societal and environmental contexts. PLO8: (Ethics): Apply ethical principles and commit to professional ethics, responsibilities, and norms of engineering practice. PLO9: (Individual work and teamwork): Function effectively as individuals and members or leaders of diverse teams and in multidisciplinary settings. PLO10: (Communication): Communicate effectively on complex engineering activities with the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and convey and receive clear instructions. PLO11: (Project management and finance): Demonstrate knowledge and understanding of engineering and management principles and apply them to their work as team members or leaders or entrepreneurs to manage projects in multidisciplinary environments. PLO12: (Life-long learning): Recognize self-awareness to engage in a lifelong learning process to reflect in the broadest context of technological change.		
35	Knowledge Profile (K) K1: A systematic, theory-based understanding of the natural sciences applicable to the discipline. K2: Conceptually-based mathematics, numerical analysis, statistics and formal aspects of computer and information science to support analysis and modeling applicable to the discipline. K3: A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline. K4: Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline much is at the forefront of the discipline.		

	K5: Knowledge that supports engineering design in a practice area. K6: Knowledge of engineering practice (technology) in the practice areas in the engineering discipline. K7: Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the professional responsibility of an engineer to public safety; the impacts of engineering activity: economic, social, cultural, environmental and sustainability. K8: Engagement with selected knowledge in the research literature of the discipline
36	Range of Complex Engineering Problem Solving (P) Complex Engineering Problems have characteristics P1 and several or all of P2 to P7: P1: Depth of knowledge required: Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6 or K8, which allows a fundamentals-based, first principles analytical approach P2: Range of conflicting requirements: Involve wide-ranging or conflicting technical, engineering and other issues P3: Depth of analysis required: Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models P4: Familiarity of issues: Involve infrequently encountered issues P5: Extent of applicable codes: Are outside problems encompassed by standards and codes of practice for professional engineering P6: Extent of stakeholder involvement and conflicting requirements: Involve diverse groups of stakeholders with widely varying needs P7: Interdependence: Are high-level problems including many component parts or sub-problems
37	Range of Complex Engineering Activities (EA) Attribute Complex activities means (engineering) activities or projects that have several or all of the following characteristics: A1: Range of resources: Involve the use of diverse resources (and for this purpose resources include people, money, equipment, materials, information and technologies). A2: Level of interaction: Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering, or other issues. A3: Innovation: Involve creative use of engineering principles and research-based knowledge in novel ways. A4: Consequences for society and the environment: Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation. A5: Familiarity: Can extend beyond previous experiences by applying principles-based approaches.
38	Code of Conduct It is strongly suggested that students keep discipline in the classroom attending on time, listening lecture attentively and participating in discussion on the subject. To get class participation grade, students MUST attend the classes of courses s/he registered for. Turn off his/her cell phone before enter in a class or participating in class tests and exams. There are activities which are considered as academic misconduct. One of them is plagiarism, which signifies the deliberate formal presentation or submission of works, phrases, texts, ideas, illustrations, or diagrams of others as one's own without proper citation. Another one is the use of unauthorized aids (including electronic devices), asking assistance or using illegal materials preparing assignments or in examinations. In addition, copying from other's work, showing your work to others and asking for answers are also considered as an academic misconduct. Penalties for involving academic misconduct include one or more of the following: a zero grade on the work produced, a failing grade in the course, suspension for one semester or more and even expulsion from the university. On the university premises or at a university-sponsored program, students must abide by the Student Code of Conduct and other Rules and Regulations of BUBT, which are available on the BUBT website at https://www.bubt.edu.bd/Home/page_details/Office_of_the_Proctor.
39	Social & Moral Values Our promises are based on the three cardinal principles: - What we do believe. - What we do practice. - What we will promote. However, students are advised to undertake the following commitments for social and moral developments.

	• To be punctual and attentive in classes; • To prioritize honesty & faith; • To ensure mutual respect; • To be always proactive; • To avoid conspiracy; • To be cooperative in learning; • To be sincere in class preparation;	• To avoid unfair means and plagiarism in exams, report writings and assignments; • To carry out assignments or keep other commitments timely; • To be motivated for asking question and encourage feedback; • Not to forget to switch-off the cellphone in a class;	• To follow the dress code and wearing ID card on campus; • To be decent on all aspects; • To be loyal and trust-worthy to the teachers and others; • Help keeping an eco-friendly environment in the campus.
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Prepared by:

Checked by:

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